System response vs. pole locations, or Detailed Interpretation of the S-plane

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1 Introduction

- One of our major goals is to tie controller specifications to a pole-zero constellation on the S-plane. For this, three major pieces are needed, as seen in figure 1.

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1. Graphical residue calculation,

   - To tie the pole-zero map to the contribution of each mode.

2. System response versus pole locations,
   - To tie poles to performance.

3. The root locus,
   - To determine how poles move around on the S-plane as controller parameters are adjusted.

Figure 1: Three needed pieces to tie controller specifications to a pole-zero constellation.

In this lecture we consider item 2, system response versus pole locations. The topics of this lecture are:

1. Basics of pole locations

   (a) Terminology for a complex pole pair (revisited)

   (b) Pole location and response characteristics

   (c) Summary: Pole location and response characteristics

2 Basics of pole locations

2.1 First Order

- We have seen that:
  - Each real pole is associated with a mode of the system response.
  - A real pole gives the terms of Eqn (1), as seen in figure 2.

$$Y(s) = \frac{3}{s+2} = \frac{C_1}{s+\alpha} \qquad \Longleftrightarrow \qquad y(t) = C_1 e^{-\alpha t} \tag{1}$$

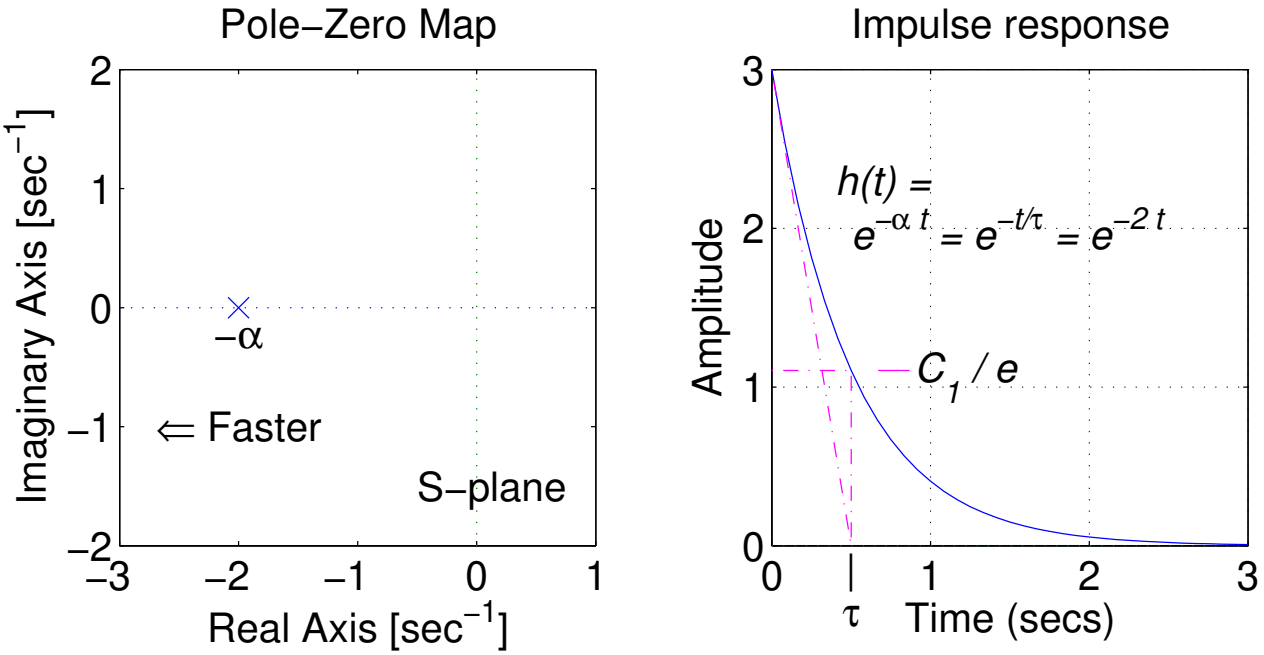


Figure 2: First order pole and impulse response.

- A real pole has these characteristics:
  - $y(t) \sim e^{-t/\tau}$ where τ [sec] = $1/\alpha$ is the time constant.
  - Further to the left indicates a faster response (smaller τ).
  - The pole-zero constellation does not show either the K_{DC} or the K_{rlg} of the transfer function or the *amplitude* of the impulse response.

2.1.1 Real Pole location and response characteristics

2.1.1.1 Example 1: Shifting poles to the left accelerates transient decay

- Two example first-order responses are shown.

  'X' Case: $\alpha = 4\ [\text{sec}^{-1}]$ $\tau = 1/4\ [\text{sec}]$

  'Δ' Case: $\alpha = 16\ [\text{sec}^{-1}]$ $\tau = 1/16\ [\text{sec}]$

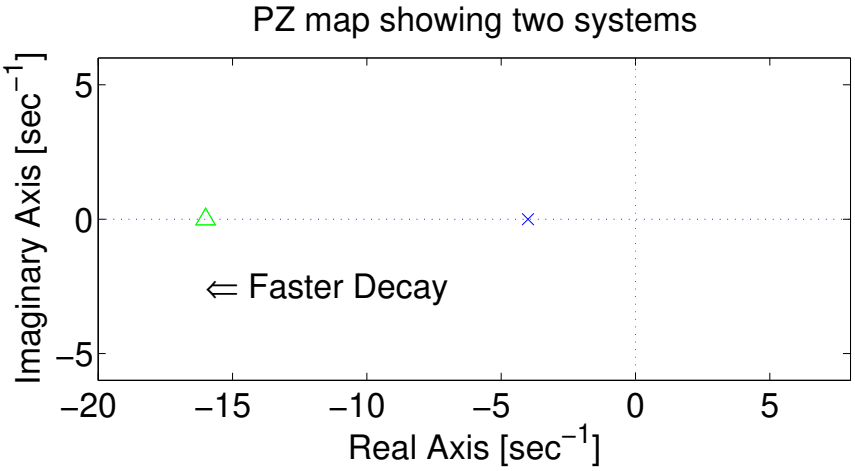


Figure 3: A change in pole location changes the decay rate and damping.

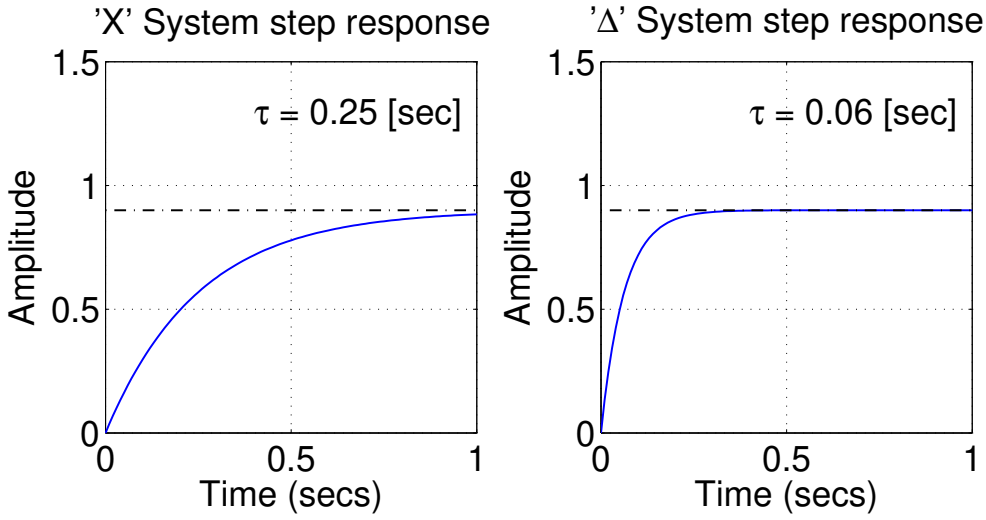


Figure 4: A change in α changes the time constant.

2.2 2nd Order

- And we have seen for second order:
  - Each complex pole pair is associated with a mode of the system response.
  - A complex pole pair gives the terms of Eqn (2), as seen in figure 5.

$$Y(s) = \frac{b_1 s + b_0}{s^2 + 2\alpha s + (\alpha^2 + \omega^2)} = \frac{2s + 14}{s^2 + 3s + 18.25} \quad (2)$$

$$\Longleftrightarrow$$

$$y(t) = A e^{-\alpha t} \cos(\omega t + \phi) = 3.4\, e^{-1.5t} \cos(4t - 54.0^o) \quad (3)$$

$$p_{1,2} = -1.5 \pm j4 = -\alpha \pm j\omega \quad (4)$$

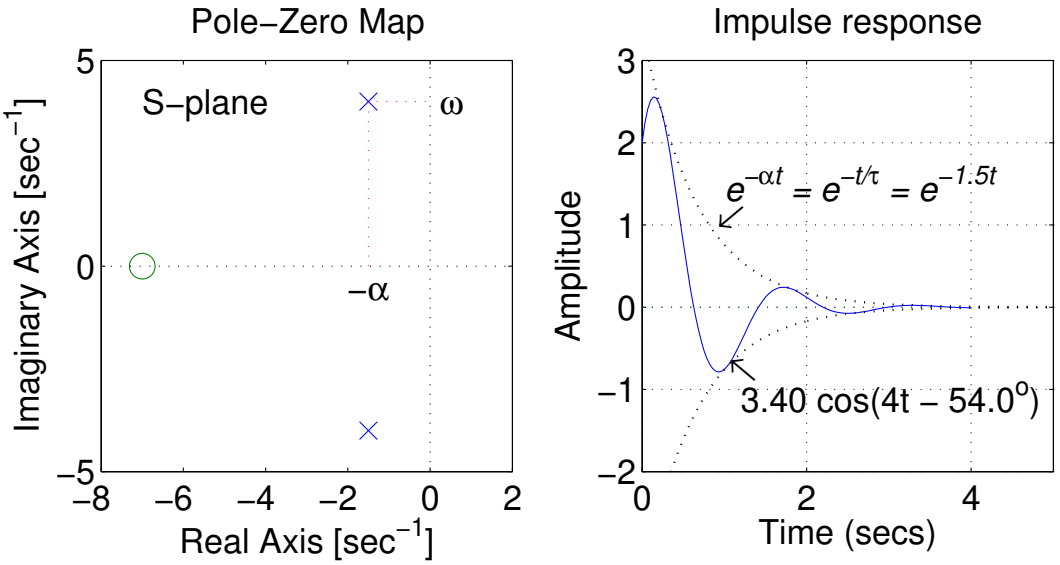
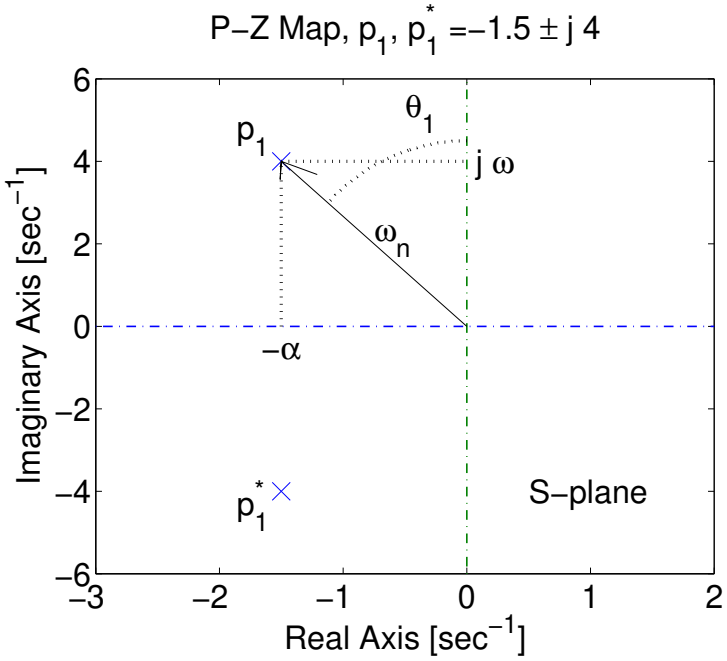


Figure 5: Second order pole and impulse response.

  - Poles determine time constants α and ω .
  - Numerator terms help determine A and ϕ, the response amplitude and phase.

2.2.1 Notation for a complex pole pair (revisited)



Example:

$$H(s) = \frac{10}{s^2 + 6s + 25}$$

Check: $H(s)$ is monic.

$$\omega_n = \sqrt{a_2} = \sqrt{25} = 5 \text{ [r/s]}$$

$$\begin{aligned} \rho &= \frac{a_1}{2\omega_n} = \frac{a_1}{10} \\ &= 0.6 \text{ [·]} \end{aligned}$$

Figure 6: Complex pole pair with ω_n and ϕ defined.

Term	Description	Given by	Units
α or σ	Decay Rate	$p_1 = -\alpha + j\omega$	$[\text{sec}^{-1}]$
ω or ω_d	Damped Nat. Freq.	$p_1 = -\alpha + j\omega$	[rad/sec]
ω_n	Natural Freq.	$\omega_n^2 = (\alpha^2 + \omega^2)$	[rad/sec]
θ	Pole Angle	$\theta = \text{atan2}(\alpha, \omega)$	[deg]
ρ or ζ	Damping Factor	$\rho = \alpha/\omega_n = \sin\theta$	Dim'less, [·]

Table 1: Factors derived from the location of a complex pole.
(Note: Franklin *et al.* often use σ, ω_d and ζ.)

Rectangular	Polar
$p_1 = \alpha + j\omega$	$p_1 = \omega_n \angle 90 + \theta$
$\alpha = \rho\,\omega_n$	$\rho = \sin(\theta)$
$\omega = \sqrt{1-\rho^2}\,\omega_n$	$\rho = \alpha / \omega_n$
$H(s) = \dfrac{\omega}{(s+\alpha)^2 + \omega^2}$	$H(s) = \dfrac{\omega_n\sqrt{1-\rho^2}}{s^2 + 2\rho\,\omega_n s + \omega_n^2}$

Table 2: The terms of table 1 relate to rectangular or polar form for the poles.

2.2.2 Complex pole location and response characteristics

2.2.2.1 Example 1: Shifting poles to the left accelerates transient decay

	α	ω	ω_n	ρ	PO
'X' Case:	1 $[\text{sec}^{-1}]$	4 [rad/sec]	4.12 [rad/sec]	0.25 [Dim'less]	44%
'Δ' Case:	4 $[\text{sec}^{-1}]$	4 [rad/sec]	5.66 [rad/sec]	0.71 [Dim'less]	4%

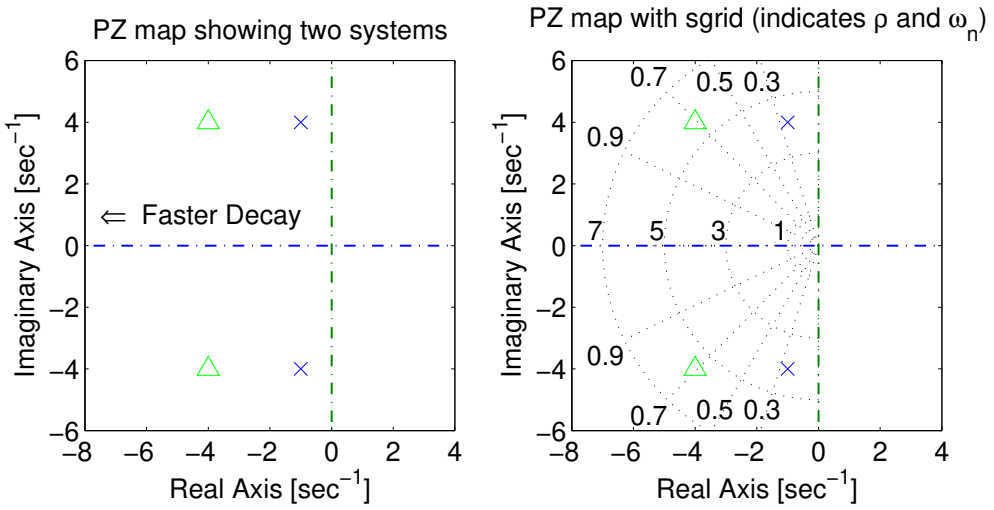


Figure 7: A change in α changes the decay rate and damping.

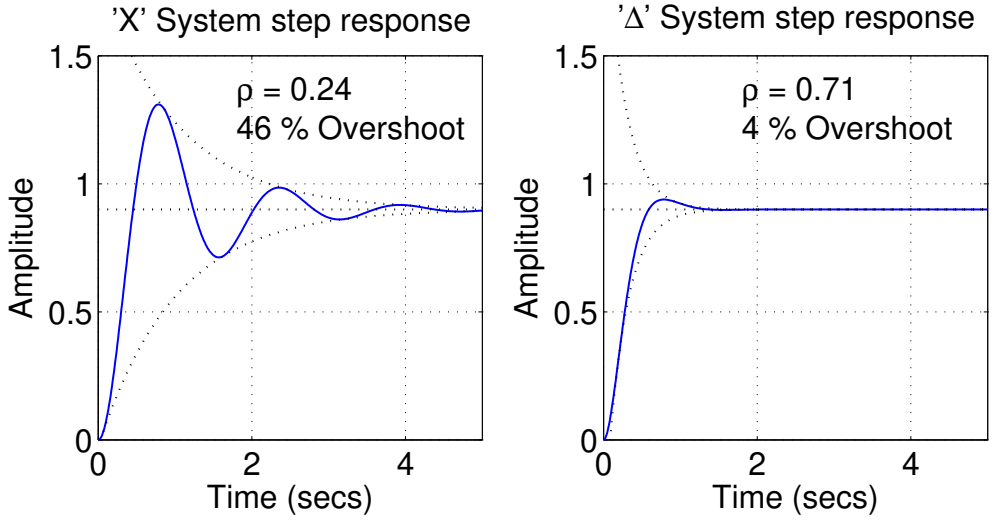


Figure 8: Step response: a change in α, ω is unchanged.

2.2.2.2 Example 2: Shifting poles out vertically increases oscillation frequency

	α	ω	ω_n	ρ	PO
'X' Case:	4 $[\text{sec}^{-1}]$	4 [rad/sec]	5.66 [rad/sec]	0.71 [Dim'less]	4%
'Δ' Case:	4 $[\text{sec}^{-1}]$	16 [rad/sec]	16.49 [rad/sec]	0.24 [Dim'less]	44%

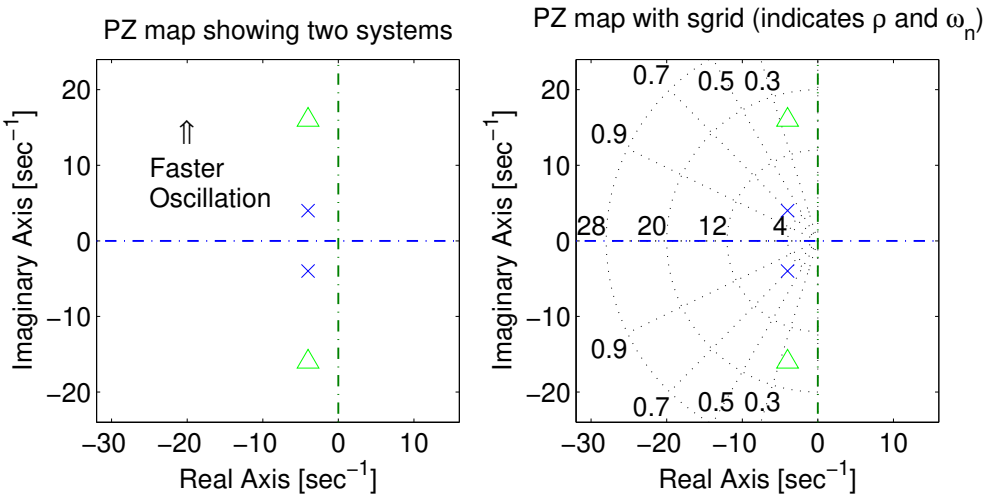


Figure 9: A change in ω changes the oscillation frequency and damping.

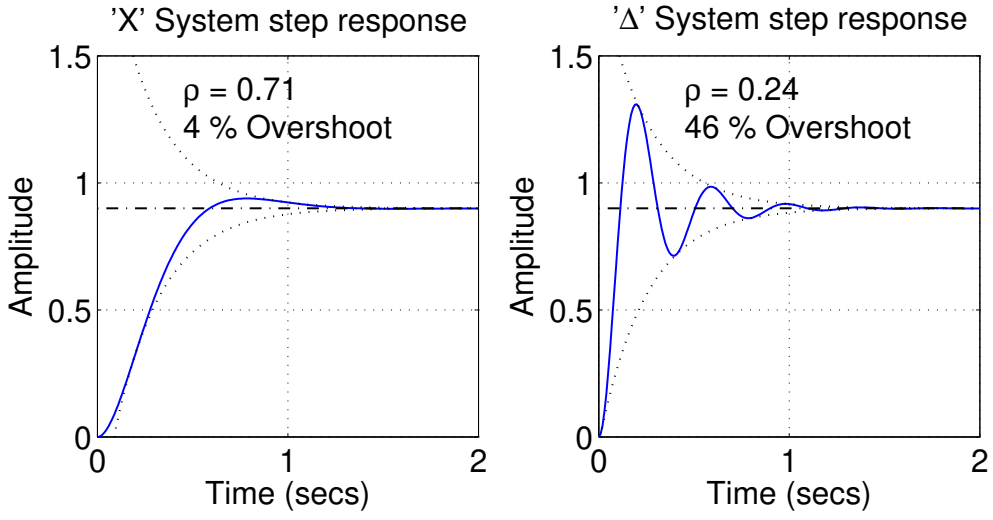


Figure 10: Step response: a change in ω.

2.2.2.3 Example 3: Shifting poles out radially rescales time

	α	ω	ω_n	ρ	PO
'X' Case:	4 $[\text{sec}^{-1}]$	4 [rad/sec]	5.66 [rad/sec]	0.71 [Dim'less]	4%
'Δ' Case:	16 $[\text{sec}^{-1}]$	16 [rad/sec]	22.63 [rad/sec]	0.71 [Dim'less]	4%

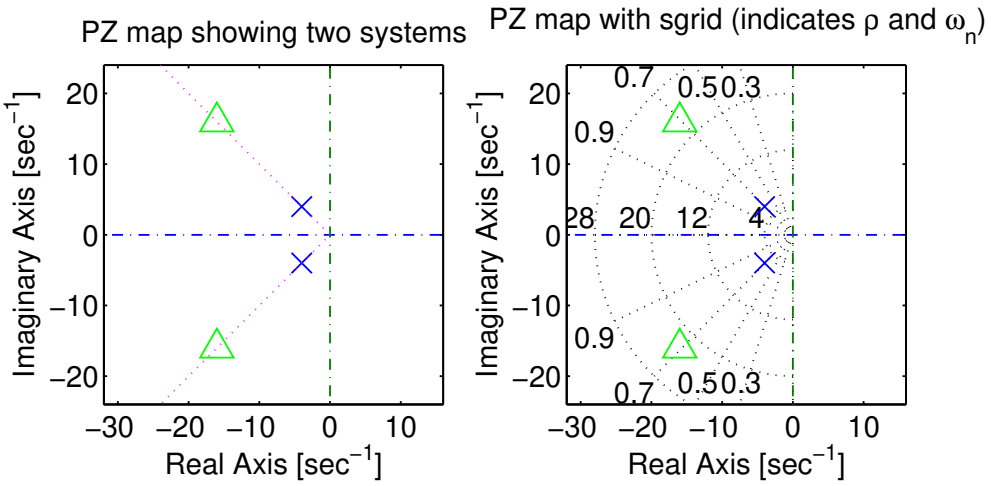


Figure 11: A radial change in pole location changes the decay rate and oscillation frequency, but not the damping.

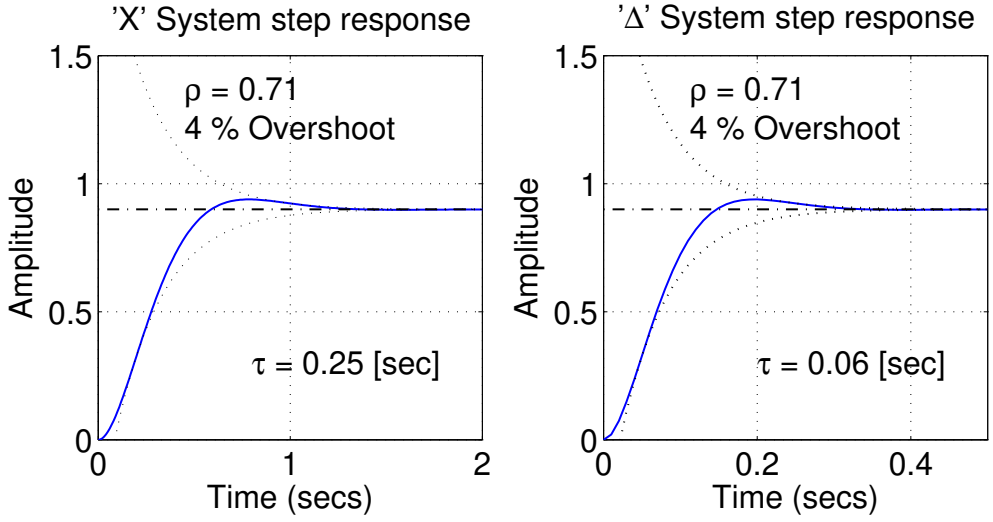


Figure 12: Maintaining ρ, time is rescaled.

Note: The S-plane has units of $[\text{sec}^{-1}]$.

2.2.3 The damping factor: ρ

- Plugging $\rho = \alpha/\omega_n$ back into the 2nd order form gives:

$$H(s) = \frac{b_0}{s^2 + a_1 s + a_0} = \frac{b_0}{s^2 + 2\rho\,\omega_n s + \omega_n^2} \tag{5}$$

giving:

$$\rho = \frac{a_1}{2\sqrt{a_0}} \quad \text{or} \quad a_1 = 2\rho\sqrt{a_2} \tag{6}$$

- ρ is defined by Eqn (6) for either real poles: $(\rho \geq 1.0)$, or a complex pole pair: $(0.0 < \rho < 1.0)$.

- A full list of what we can learn from ρ is seen in table 3.

ρ value	System characteristic
$\rho < 0$	Unstable
$\rho = 0$	Marginally Stable (poles on imaginary axis)
$0 < \rho < 1$	Stable Complex Pole Pair
$0.5 < \rho \leq 1.0$	Typical value for a well designed controller
$\rho = 1$	Critical Damping (repeated real poles)
$\rho > 1$	Over Damped (two separate real poles)

Table 3: Ranges of damping factor.

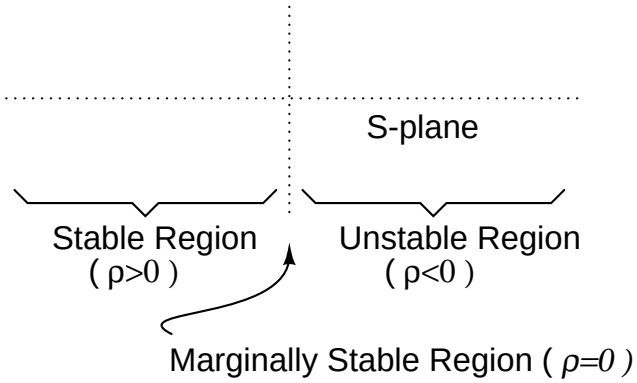


Figure 13: Damping factor and stability.

3 Higher order systems: dominant mode

- The response of a higher order system is the superposition of the responses of the modes:
  - 1st order modes;
  - 2nd order modes.
- Typically (not quite always) one mode will dominate. In the example of figure 14, the slower second-order mode dominates.

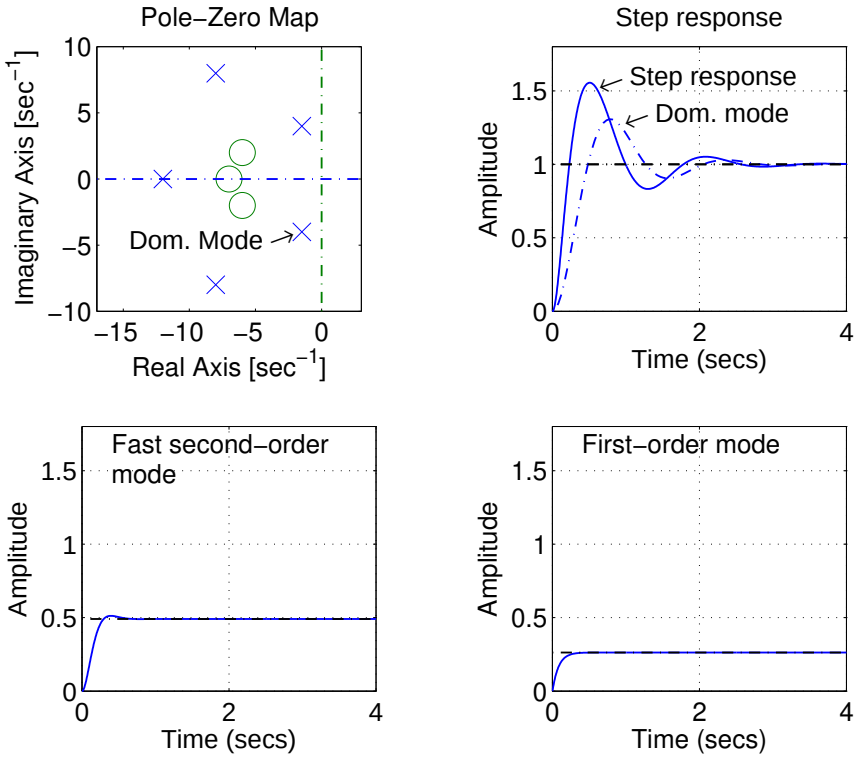


Figure 14: Example high-order system showing dominant mode in step response.

- Recall: a goal of studying the S-plane and pole-zero constellation is to tie controller specifications to pole locations.
  - The dominant mode dominates the response.
  - The root-locus design method operates by placing the ***dominant*** poles in the desired region of the S-plane

    (Other poles can be anywhere in stable region).

Dominant mode (continued)

- As the figure 14 "Step response" plot shows, the dominant mode is only an approximation to the true step response.
- For high-order systems $(n \geq 3)$, design control by:
  1. Use *dominant mode* concept and approximation for initial design; then
  2. Fine tune controller based on actual (high order) response.

3.1 Which mode is the dominant mode ?

- The dominant mode is:
  - A slow mode, with a
  - Large residue (a large contribution to the output)
- On the S-plane this means that the dominant mode is (almost always):
  - The slowest mode (other modes die out faster).
- Exceptionally, one must also consider the size of the residue, the pole of the dominant mode should be:
  - Near to other poles; for example, $s = 0$ pole from step input
    (Graphical method: being near other poles gives a large residue).
  - Far from zeros
    (Graphical method: being far from zeros gives a large residue).

Dominant mode (continued)

- A second example is presented in figure 15.
- Which mode is dominant in figure 15 ? Student Exercise

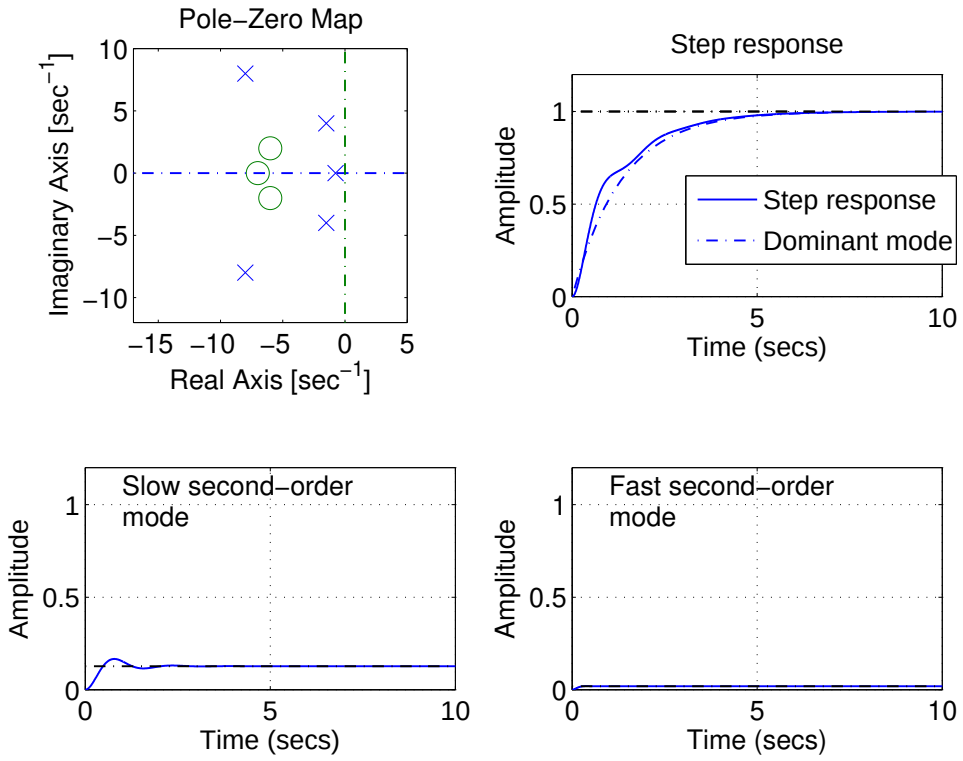


Figure 15: Second example showing dominant mode in step response.

- Why are the contributions of the second-order terms so small in figure 15 ? (Hint: consider the graphical residue method.) Student Exercise

Dominant mode (continued)

- A third example is presented in figure 16.
- Which mode is dominant in figure 16 ? Student Exercise

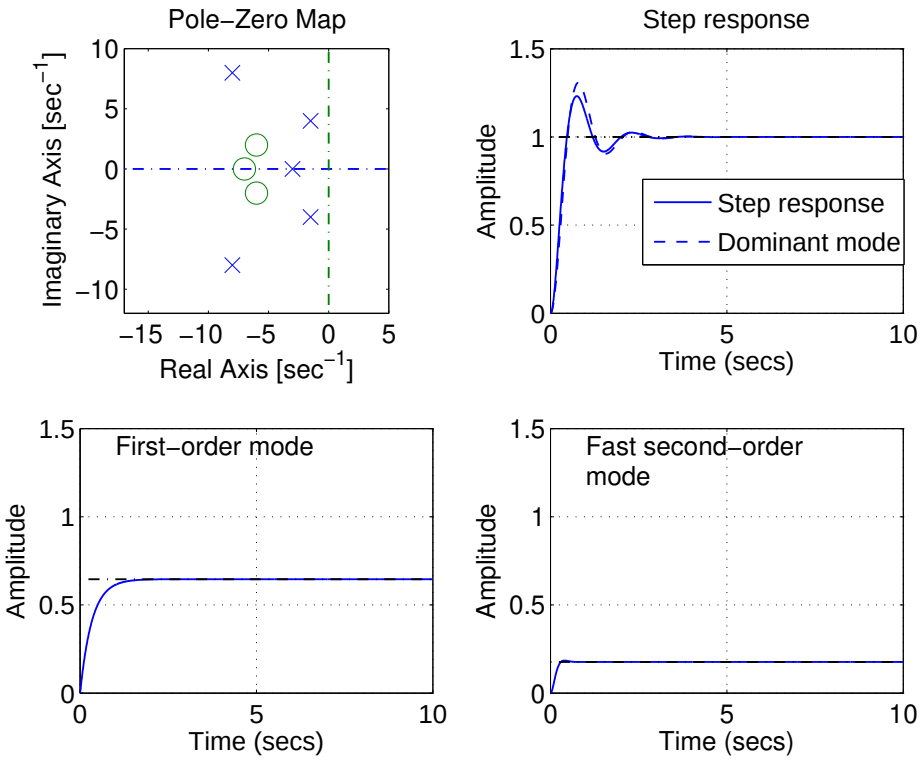


Figure 16: Third example showing dominant mode in step response.

- Why is the first order mode nearly invisible in the step response, even though the residue of the first order mode is pretty large ? Student Exercise

4 Summary: Pole location and response characteristics

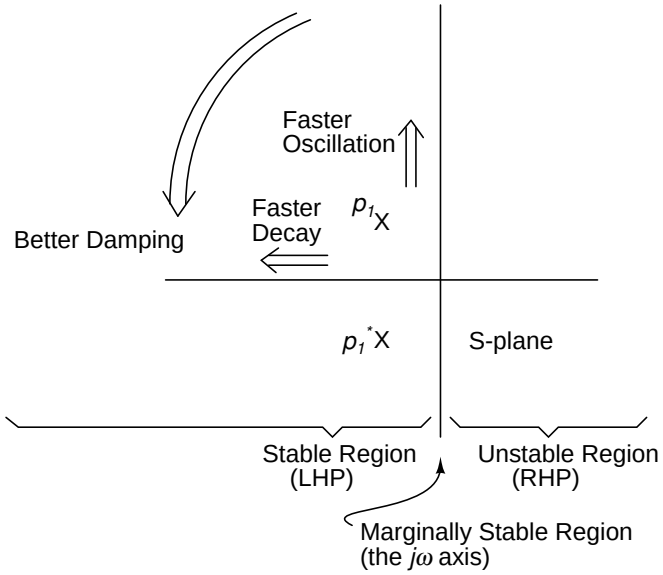


Figure 17: Decay rate, oscillation freq. and damping depend upon pole location.

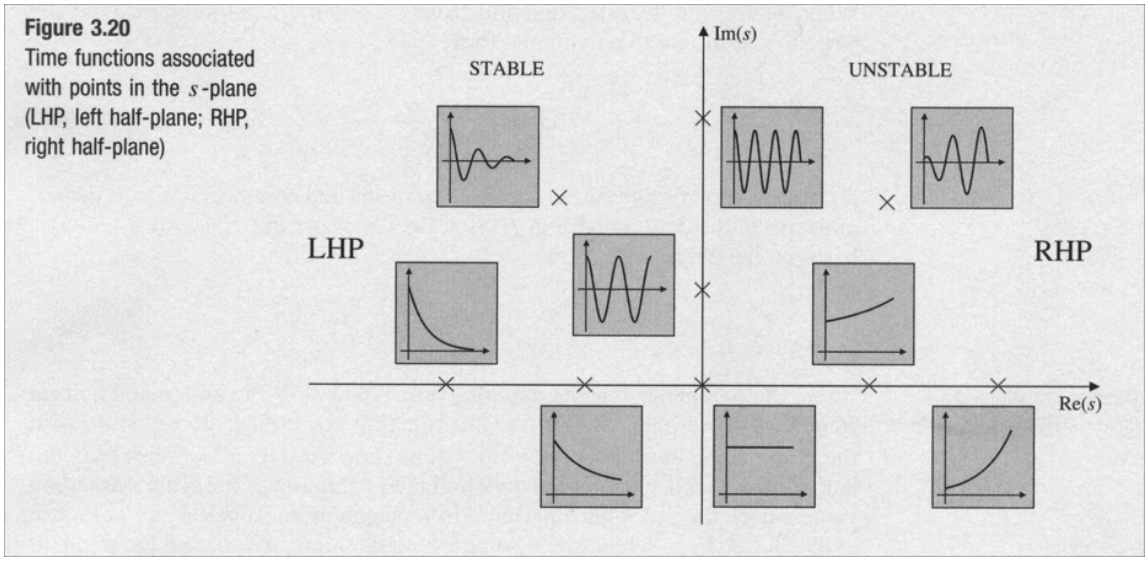


Figure 18: S-plane and impulse responses.

Summary (continued)

- Pole locations for a range of example systems are shown in figure 19.

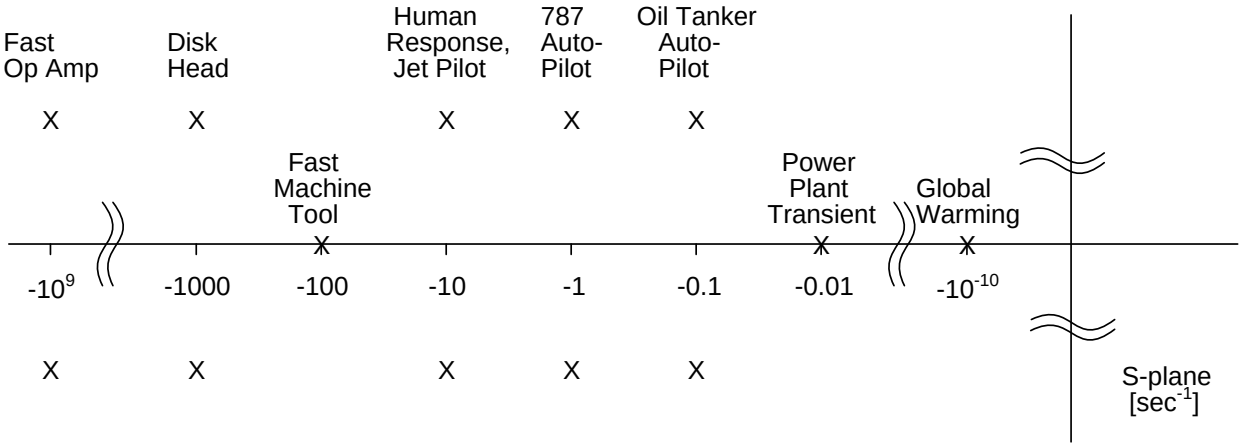


Figure 19: Example pole locations on logarithmic scale.

- Some systems most likely have a first-order dominant mode, while others most likely have a second-order dominant mode.

Student Exercises

1. What are the time constants for the systems in figure 19 ?
2. Give another example system for some or all of the time constants in figure 19.